Databases – Beginner Notes

1. Introduction to Databases

A database is an organized collection of data that can be stored, accessed, updated, and managed efficiently. Databases are used by websites, mobile applications, banks, hospitals, schools, companies, and many other systems to keep information available and consistent.

A Database Management System (DBMS) is software that helps users and applications create, store, retrieve, update, and manage data in a database. Examples of database systems include MySQL, PostgreSQL, Oracle Database, Microsoft SQL Server, and SQLite.

2. Why Databases Are Used

- Store large amounts of information in an organized way
- Search and retrieve information efficiently
- Add, update, and delete records
- Control access to information
- Reduce unnecessary duplication of data
- Support multiple users and applications
- Maintain data consistency and reliability

3. Types of Databases

Relational Databases: Store data in tables made of rows and columns. They commonly use SQL for querying and managing data. Examples include MySQL, PostgreSQL, Oracle Database, and SQL Server.

NoSQL Databases: Use data models other than the traditional relational table model. Common categories include document, key-value, column-family, and graph databases. They can be useful for certain large-scale or flexible-data applications.

Hierarchical Databases: Organize data in a tree-like parent-child structure.

Graph Databases: Represent entities as nodes and relationships as connections, making them useful for applications where relationships are especially important.

4. Basic Database Terms

Table: A structure used to organize related data in rows and columns.

Row/Record: A single entry in a table.

Column/Field: A category or attribute describing data stored in a table.

Primary Key: A column or combination of columns that uniquely identifies a row.

Foreign Key: A field that references a key in another table and helps create relationships.

Query: A request to retrieve or manipulate data.

5. SQL – Structured Query Language

SQL is widely used with relational databases. It allows users to create database structures, insert data, retrieve information, modify records, and control access.

Common SQL commands include:

SELECT – retrieves data from one or more tables.

INSERT – adds new records.

UPDATE – changes existing records.

DELETE – removes records.

CREATE – creates database objects such as tables.

ALTER – changes the structure of an existing object.

DROP – removes a database object.

6. Example of a Simple Table

Consider a Students table with the columns Student_ID, Name, Course, and Marks. Each row represents one student. Student_ID can be used as a primary key because each student record should have a unique identifier.

A simple SQL query such as **SELECT Name, Marks FROM Students;** can be used to retrieve the names and marks of students. A condition can be added with a WHERE clause to retrieve only records that satisfy a particular requirement.

7. Relationships Between Tables

Relational databases can connect tables using keys. A **one-to-one** relationship links one record to one related record. A **one-to-many** relationship occurs when one record is related to many records. A **many-to-many** relationship occurs when many records in one table can be related to many records in another table, usually through an intermediate table.

8. Database Normalization

Normalization is a technique used to organize relational data and reduce unnecessary duplication. It divides information into related tables and establishes appropriate relationships between them. Common normal forms include First Normal Form (1NF), Second Normal Form (2NF), and Third Normal Form (3NF).

9. Transactions and ACID

A database transaction is a logical unit of work that should be completed reliably. The ACID properties are Atomicity, Consistency, Isolation, and Durability. Together, these properties help maintain reliable data when transactions are processed.

10. Database Security

Database security protects information from unauthorized access, accidental changes, and other threats. Important practices include strong authentication, role-based access, least-privilege permissions, encryption where appropriate, secure backups, software updates, and monitoring of database activity.

11. Database Backup and Recovery

Backups are copies of data that can be used to recover information after accidental deletion, hardware failure, software problems, or other incidents. A good backup strategy considers how often backups are taken, where they are stored, how long they are retained, and whether recovery procedures have been tested.

12. DBMS Advantages

- Centralized and organized data management
- Faster data retrieval and processing
- Better control over access and permissions
- Data integrity and consistency features
- Backup and recovery capabilities
- Support for multiple applications and users

13. Databases in Real Life

Banks use databases for customer accounts and transactions. Hospitals store patient and appointment information. Schools maintain student, course, and examination records. Online shopping systems store product, customer, order, and payment-related information. Social applications also use databases to manage accounts, content, and relationships.

14. Basic Database Learning Path

1. Learn tables, rows, columns, keys, and relationships.
2. Learn basic SQL commands such as SELECT, INSERT, UPDATE, and DELETE.
3. Practice WHERE, ORDER BY, GROUP BY, and JOIN operations.
4. Learn database design and normalization.
5. Understand transactions, indexes, security, and backups.
6. Practice with a database such as MySQL, PostgreSQL, or SQLite.

15. Conclusion

Databases are a fundamental part of modern software and information systems. Understanding database concepts, relational tables, keys, SQL, relationships, normalization, transactions, security, and backups provides a strong foundation for programming, web development, data analysis, and IT careers.